

Layered parallax: research and release review

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Decision: approve the corrected implementation

The uncommitted version was not ready to ship unchanged. Review found reversed depth motion, two light-theme hover contrast failures, a narrow-phone gap above the artwork, and avoidable image weight. Those issues are corrected. All 66 browser tests and 94 unit tests pass; lint, formatting, typecheck, production build, dependency audit, and the production SBOM gate also pass.

This review covers the portfolio hero and its interaction with the accompanying light-theme changes. It uses primary browser documentation and W3C guidance, direct code inspection, rendered browser and desktop screenshots, and local measurements. It is a release decision for this repository, not a general endorsement of parallax or a claim of site-wide WCAG conformance.

The depth model that matters

For a non-sticky layer, screen position equals its document position minus the scroll offset plus its animated translation. A positive downward translation therefore cancels some upward scrolling. Distant layers need MORE downward compensation to move LESS on screen. The original implementation did the reverse. This reasoning agrees with [Chrome's explanation of parallax geometry](#).

At a 1440px viewport, final compensation ranges are 144px for the backdrop, 86.4px for the technology layer, and 36px for the skyline. In a measured 350px scroll, their upward viewport movements were approximately 300.7, 324.0, and 344.4px respectively. The common header-height change affects absolute travel, but the depth ordering remains correct. Text and controls follow normal document scrolling.

The native view timeline now runs through the complete exit range. Explicit zero timeline inset prevents the site's anchor-navigation scroll padding from shifting animation progress. The [CSSWG working draft](#) defines these range and inset semantics; it remains a work in progress.

Measured upward screen travel after a 350px scroll

Backdrop		301px
Technology		324px
Skyline		344px

What changed and why it passes

Finding	Corrected behavior
Foreground moved slowest on screen	Backdrop moves slowest; skyline moves fastest. Tests check viewport movement as well as transforms.
Light hero hover text measured 1.28:1 and 1.25:1	Theme-specific hover surfaces now measure 9.67:1 and 13.99:1. Focus remains visible.
Positive background offsets uncovered a strip at 320px	Mobile offsets are capped at the top edge; rendered review confirms continuous coverage.
Animation ended at 65% of the exit	The effect continues through 100%; scrolling back restores the starting pose.
Large layer payloads	Responsive AVIF images are preferred, with WebP fallbacks and no new dependency.

The artwork remains three decorative, nonfocusable layers behind real text. Its container is hidden from assistive technology, and pointer events pass through. No JavaScript scroll loop, animation package, or forced layer-promotion hint was added.

Nonessential interaction motion can cause vestibular discomfort. [W3C's SC 2.3.3 guidance](#) explicitly discusses parallax; this criterion is AAA. The current static-first CSS follows [W3C technique C39](#): animation requires an OS no-preference setting and a supported timeline. Phones at 600px or narrower remain static. Both directions of live OS preference changes are tested. The internal data-motion attribute mirrors OS state; the app does not expose a separate motion toggle.

Chromium and WebKit animate at supported desktop/tablet widths. Installed Firefox defaults to static artwork; a dedicated Firefox test also disables timeline support explicitly and verifies navigation still works. Feature queries do not guarantee bug-free browser implementations: [Safari 26.5 release notes](#) include timeline boundary fixes, which supports testing actual engines and scroll reversal.

Loading cost and measurement limits

All figures below sum the three selected source responses, in decimal kilobytes. They compare the reviewed uncommitted layers with the corrected assets, not with the previous single-image Next.js delivery.

Theme / crop	Original WebP	Preferred AVIF	Reduction
Light / wide	2,568.4 KB	429.5 KB	83.3%
Dark / wide	1,510.3 KB	341.0 KB	77.4%
Light / mobile	1,136.9 KB	218.9 KB	80.7%
Dark / mobile	747.4 KB	191.2 KB	74.4%

Crops are 3200px wide for desktop, 2160px for tablet, and 1200px for phone. AVIF uses quality 50 / effort 3; fallback WebP uses quality 78 / alpha quality 95 / effort 5. The browser selects one format and the active theme/crop. Chromium request records confirmed three selected AVIF files and no simultaneous fallback downloads. [MDN's image-set documentation](#) describes format-aware selection.

Three fresh light-theme Chromium contexts per width used a local HTTPS production preview, cold browser cache, 6 Mbps download, 150ms latency, and 4x CPU throttling. LCP samples were 1.644s, 1.472s, 1.520s at 390px and 1.812s, 1.740s, 1.756s at 1440px. All six measured CLS values were zero. These are local synthetic samples, not production percentiles or physical-device benchmarks. An intercepted-response comparison was discarded because fulfillment bypassed the emulated transfer path.

CSS backgrounds remain discoverable after CSS applies. [Google's LCP guidance](#) recommends selective preload for a critical background image, but this app's persisted theme can differ from OS theme. A blind OS-based preload could fetch the wrong artwork. With the reduced payload and passing local smoke results, no broad preload was added. Production Web Vitals should determine whether matching the resolved theme in an early preload is worthwhile.

Native transform animation is an appropriate implementation choice; [Chrome's performance case study](#) demonstrates its advantages under main-thread load. That study does not prove this scene's GPU memory use or frame rate. Blending, raster decoding, very high-density screens, and lower-powered hardware remain costs to monitor. No universal 60fps claim is made.

Validation and maintenance record

Automated gates passed: 94 unit tests across 29 files; 66 production browser tests across Chromium, WebKit, Firefox fallback, phone/tablet profiles, theme preferences, and reduced motion. Coverage is 89.27% statements, 84.00% branches, 90.42% functions, and 89.51% lines. Biome, strict TypeScript, Next.js production build, dependency audit, SBOM generation, and whitespace checks pass. No dependencies changed.

Rendered review included both themes at 320, 390, 768, 1440, and 2560px, plus WebKit tablet/desktop and Firefox fallback. Top, middle, and late scroll positions were measured; selected screenshots were inspected for crop, seams, text readability, hover/focus, and layout. A wider automated sweep covered 320 through 2560px and breakpoint edges with no horizontal overflow. Headed Chromium and real Flameshot desktop captures supplement browser screenshots. Screenshot review was sampled, not exhaustive across every possible display.

Retain three planes, static-first CSS, readable foreground overlays, and native anchor navigation. When changing the motion, test screen displacement rather than assuming larger transform values mean faster foreground travel. When replacing artwork, check painted-image coverage as well as element bounds. Keep the selected theme, image format, screenshot links, and capture provenance aligned.

Sources and dates

Sources were accessed September 8, 2026. Chrome, W3C, WebKit, and Mozilla documentation supplied the consequential claims; local code and measurements supplied implementation findings. Targeted follow-ups resolved geometry, timeline insets, motion preferences, format selection, and image-loading tradeoffs. Research stopped once remaining questions required runtime measurements rather than more literature.

- Google Chrome, Paul Lewis and Robert Flack: [Performant Parallaxing](#), updated December 2, 2016. Geometry remains useful; its historical browser workarounds were not adopted.
- CSSWG: [Scroll-driven Animations Module Level 1](#), Editor's Draft May 14, 2026.
- W3C WAI: [Animation from Interactions](#), updated September 16, 2025; [C39](#), updated January 12, 2026.
- WebKit, Jen Simmons: [WebKit Features for Safari 26.5](#), May 11, 2026.
- Mozilla MDN: [image-set\(\)](#), live reference, accessed September 8, 2026.
- Google web.dev, Philip Walton and Barry Pollard: [Optimize Largest Contentful Paint](#), updated March 31, 2025.
- Google Chrome: [A case study on scroll-driven animations performance](#), updated July 12, 2023.